THE CHARACTERISTIC POLYNOMIAL OF A FIELD ELEMENT AS A LINEAR MAP IS A POWER OF ITS MINIMAL POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This lemma feeds lemma-trace-and-norm-from-minimal-polynomial (source lines 2642-2658) and the trace-pairing separability criterion (delivered L48333).

Editorial note. The label lemma-characteristic-vs-minimal-polynomial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-degree-minimal-polynomial (deg of the minimal polynomial equals the subextension degree) and lemma-multiplicativity-degrees (delivered card L48338) for e deg(P) = [L : K]; Cayley-Hamilton; minimal polynomial divides the characteristic polynomial

Source reference. Original reference 09GN: sources/stacks-fields/pinned-original-source.tex, source lines 1087–1087. Prerequisite pointer invoked at 2626; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09G9: sources/stacks-fields/pinned-original-source.tex, source lines 759–759. Prerequisite pointer invoked at 2627; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a finite extension of fields. Let $\alpha \in L$ and let P be the minimal polynomial of α over K. Then the characteristic polynomial of the K-linear map $\alpha : L \to L$ is equal to P^e with $e \deg(P) = [L : K]$.*

Proof. Choose a basis $\beta_1, \ldots, \beta_e$ of L over $K(\alpha)$. Then e satisfies $e \deg(P) = [L : K]$ by Lemmas 09GN and 09G9. Then we see that $L = \bigoplus K(\alpha)\beta_i$ is a direct sum decomposition into α-invariant subspaces hence the characteristic polynomial of $\alpha : L \to L$ is equal to the characteristic polynomial of $\alpha : K(\alpha) \to K(\alpha)$ to the power e.

To finish the proof we may assume that $L = K(\alpha)$. In this case by Cayley-Hamilton we see that α is a root of the characteristic polynomial. And since the characteristic polynomial has the same degree as the minimal polynomial, we find that equality holds. □

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